

HTML DOM Document

getElementsByClassName()

The `getElementsByClassName()` method returns a collection of elements with a specified class name(s).

The `getElementsByClassName()` method returns an [HTMLCollection](#).

HTMLCollection

An HTMLCollection is an array-like collection (list) of HTML elements.

The length Property returns the number of elements in the collection.

The elements can be accessed by index (starts at 0).

An HTMLCollection is live. It is automatically updated when the document is changed.

```
<html>
  <head>
    <style>
      .example {
        border: 1px solid black;
        padding: 8px;
      }
    </style>
  </head>
  <body>
    <h1>The Document Object</h1>
    <h2>The getElementsByClassName() Method</h2>
    <p> Change the background color of all elements with
      class="example":</p>
    <div class="example">A div with class="example"</div>
    <br>
    <div class="example"> A div with class="example" </div>
    <p class="example"> A p element with class="example". </p>
    <p>A <span class="example">span</span> element with
      class="example".</p>
    <script>
      const collection = document.getElementsByClassName("example");
      for (let i = 0; i < collection.length; i++) {
        collection[i].style.backgroundColor = "red";
      }
    </script>
  </body>
</html>
```

Example 2

```
<html>
<head>
  <style>
    .example {
      border: 1px solid black;
      padding: 8px;
    }
  </style>
</head>
<body>

  <h1>The Document Object</h1>
  <h2>The getElementsByClassName() Method</h2>

  <p>Change the background color of all elements with
  class="example":</p>

  <div class="example"> A div with class="example" </div>
  <br>
  <div class="example"> A div with class="example" </div>

  <p class="example"> A p element with class="example". </p>

  <p> A <span class="example"> span </span> element with
  class="example".</p>
  <input type="text" id="t">
  <button type="button" onclick="myFunction(t.value)">change color
  </button>

  <script>
    function myFunction(tt) {
      const collection = document.getElementsByClassName("example");
      for (let i = 0; i < collection.length; i++) {
        collection[i].style.backgroundColor = tt;
      }
    }
  </script>
</body>
</html>
```

Example 3

```
<html>
  <body>
    <h1>The Document Object</h1>
    <h2>The getElementsByClassName() Method</h2>
    <p>Change the text of the first element with class="example":</p>
    <div class="example">Element1</div>
    <div class="example">Element2</div>
    <script>
      const collection = document.getElementsByClassName("example");
      collection[0].innerHTML = "Hello World!";
    </script>
  </body>
</html>
```

JavaScript Cookies

What are Cookies?

Cookies are data, stored in small text files, on your computer.

When a web server has sent a web page to a browser, the connection is shut down, and the server forgets everything about the user.

Cookies were invented to solve the problem "how to remember information about the user":

- When a user visits a web page, his/her name can be stored in a cookie.
- Next time the user visits the page, the cookie "remembers" his/her name.

Cookies are saved in name-value pairs like:

```
username = John Doe
```

When a browser requests a web page from a server, cookies belonging to the page are added to the request. This way the server gets the necessary data to "remember" information about users.

Create a Cookie with JavaScript

JavaScript can create, read, and delete cookies with the **document.cookie** property.

With JavaScript, a cookie can be created like this:

```
document.cookie = "username=John Doe";
```

You can also add an expiry date (in UTC time). By default, the cookie is deleted when the browser is closed:

```
document.cookie = "username=John Doe; expires=Thu, 18 Dec 2013 12:00:00 UTC";
```

With a path parameter, you can tell the browser what path the cookie belongs to. By default, the cookie belongs to the current page.

```
document.cookie = "username=John Doe; expires=Thu, 18 Dec 2013 12:00:00 UTC; path="/;
```

Read a Cookie with JavaScript

With JavaScript, cookies can be read like this:

```
let x = document.cookie;
```

Change a Cookie with JavaScript

With JavaScript, you can change a cookie the same way as you create it:

```
document.cookie = "username=John Smith; expires=Thu, 18 Dec 2013 12:00:00 UTC; path="/;
```

The old cookie is overwritten.

Delete a Cookie with JavaScript

Deleting a cookie is very simple.

You don't have to specify a cookie value when you delete a cookie.

Just set the expires parameter to a past date:

```
document.cookie = "username=; expires=Thu, 01 Jan 1970 00:00:00 UTC; path="/;
```

You should define the cookie path to ensure that you delete the right cookie.

Some browsers will not let you delete a cookie if you don't specify the path.

JavaScript Cookie Example

In the example to follow, we will create a cookie that stores the name of a visitor.

The first time a visitor arrives to the web page, he/she will be asked to fill in his/her name. The name is then stored in a cookie.

The next time the visitor arrives at the same page, he/she will get a welcome message.

For the example we will create 3 JavaScript functions:

1. A function to set a cookie value
2. A function to get a cookie value
3. A function to check a cookie value

Example 4

```
<html>
<script>
    function setCookies(cname,cvalue,ExpDay) {
        const d=new Date();
        d.setTime(d.getTime()+ (ExpDay*24*60*60*1000));
        document.cookie= cname
        + "=" + cvalue + "; expires=" + d.toDateString() + "; path=/"
    }

    function getCookie(cname) {
        const cookies=document.cookie.split("; ");
        //document.getElementById("cc").innerHTML=cookies;

        for (let i=0;i<cookies.length;i++){
            //document.getElementById("cc").innerHTML=cookies[i];
            if (cookies[i].startsWith(cname + "="))
                return cookies[i].split("=")[1];
        }
        return "";
    }

    function checkCookie() {
        let user=getCookie("Uname");
        if ( user !== "") {
            alert("welcome " + user);
        }
        else
        {
            user= prompt("enter user name", "");
            if (user !== "" && user !== null)
                {setCookies("Uname",user,30); }
        }
    }
</script>
<body onload="checkCookie()">
<p id="cc"></p>
</body>
</html>
```